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# Topología inicial

**Definición 1 (Topología inicial).** Sean $X \neq \varnothing$, $\{(Y_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in I}$ un conjunto de espacios topológicos y $\mathcal{F} = \{f_\alpha \colon X \to Y_\alpha\}_{\alpha \in I}$, $\sigma(X, \mathcal{F})$ es la topología inicial en $X$ inducida por $\mathcal{F} \iff$ es la topología menos fina que hace continuas todas las $f_\alpha \in \mathcal{F}$.

Como la intersección de topologías es una topología, tenemos que

$$\sigma(X, \mathcal{F}) := \bigcap \{\mathcal{T} \text{ topología en } X : \forall \alpha \in I : f_\alpha \colon (X, \mathcal{T}) \to (Y_\alpha, \mathcal{T}_\alpha) \text{ es continua}\}.$$

## Referenciado en

- `teo-abiertos-topologia-inicial`
- `prop-base-topologia-inicial`
- `topologia-debil`
- `prop-subbase-topologia-inicial`
- `prop-carac-convergencia-topologia-inicial`
- `lem-carac-continuidad-topologia-inicial`

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